Student's t-Distribution

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Introduction

The t-distribution (also called Student's t, after W. S. Gosset's pseudonym) arises as the ratio of a standard normal to the scaled square root of an independent chi-squared. It is the reference distribution for t-tests and for small-sample confidence intervals for a normal mean with unknown variance.

Prerequisites

Normal distribution, chi-squared distribution.

Theory

If $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_\nu$ are independent, then

$$T = \frac{Z}{\sqrt{V/\nu}} \sim t_\nu,$$

the t-distribution with ν **degrees of freedom**.

PDF:

$$f(t) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}.$$

Properties:

- Symmetric about 0.
- $E[T] = 0$ for $\nu > 1$; $\mathrm{Var}(T) = \nu/(\nu-2)$ for $\nu > 2$.
- Heavier tails than normal; excess kurtosis $6/(\nu-4)$ for $\nu > 4$.
- As $\nu \to \infty$, $t_\nu \to \mathcal{N}(0,1)$.

Pivotal quantity: for iid normal $X_1, \ldots, X_n$,

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}.$$

This is the basis of the one-sample t-test and its confidence interval.

Assumptions

Normal underlying data (or large enough n for CLT to make the approximation reasonable).

R Implementation

```r
# PDF comparison
x <- seq(-5, 5, length.out = 400)
plot(x, dnorm(x), type = "l", col = "black", lwd = 2,
     main = "t with 3, 10, 30 df vs N(0,1)", ylab = "f(x)")
lines(x, dt(x, df = 3),  col = "#F4A261", lwd = 2)
lines(x, dt(x, df = 10), col = "#6A4C93", lwd = 2)
lines(x, dt(x, df = 30), col = "#2A9D8F", lwd = 2)
legend("topright", c("normal", "t(3)", "t(10)", "t(30)"),
       col = c("black", "#F4A261", "#6A4C93", "#2A9D8F"), lwd = 2)

# Verify construction
n <- 1e5; nu <- 5
Z <- rnorm(n); V <- rchisq(n, df = nu)
T_sim <- Z / sqrt(V / nu)
c(emp_mean = mean(T_sim), theoretical = 0,
  emp_var  = var(T_sim),  theoretical_var = nu / (nu - 2))

# Small-sample CI for a mean
set.seed(2026)
X <- rnorm(15, mean = 10, sd = 3)
t_crit <- qt(0.975, df = length(X) - 1)
mean(X) + c(-1, 1) * t_crit * sd(X) / sqrt(length(X))
```

Output & Results

```
emp_mean theoretical  emp_var theoretical_var
  -0.003           0    1.665           1.667

[1]  8.76  12.06     # 95% CI using t_14
```

Empirical moments match theoretical at $\nu = 5$, and the small-sample CI properly uses t_{14} critical values.

Interpretation

Reporting a t-interval or t-test with "$t(n-1) = \ldots$" in a manuscript is reporting a t-distribution-based inference. The df equals $n-1$ for the one-sample t-test; Welch's two-sample t-test uses Satterthwaite df that may be fractional.

Practical Tips

- For $\nu \geq 30$ the t is nearly indistinguishable from the normal; for $\nu \leq 10$ the tail differences are substantial.
- The t-distribution is the one-parameter generalisation of the normal that allows heavier tails.
- Robust regression models sometimes use a t-likelihood to achieve outlier resistance.
- Non-central t (with non-centrality μ) governs power calculations for t-tests.
- For very heavy tails ($\nu \leq 2$), variance is infinite; mean exists only for $\nu > 1$.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**

- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots